From the approximate equivalent circuit, $I_2' = \frac{V}{\sqrt{\left(R_1+\frac{R_2'}{s}\right)^2+(X_1+X_2')^2}} = \frac{V}{(X_1+X_2')}\sin\emptyset$

$$\text{where } \sin\emptyset = \frac{X_1+X_2'}{\sqrt{\left(R_1+\frac{R_2'}{s}\right)^2+(X_1+X_2')^2}}$$

If the leakage reactances X_1 and X'_2 are assumed to remain constant regardless of load, and the applied voltage V is constant, the above equation represents the polar equation of a circle with diameter $\frac{V}{X_1+X_2'}$ By changing the load R_L (where $R_L=R_2'\frac{(1-S)}{s}$) and $\emptyset_1$, the value of the current I_2' changes. The locus of the current, however, lies on a circle (Figure l).

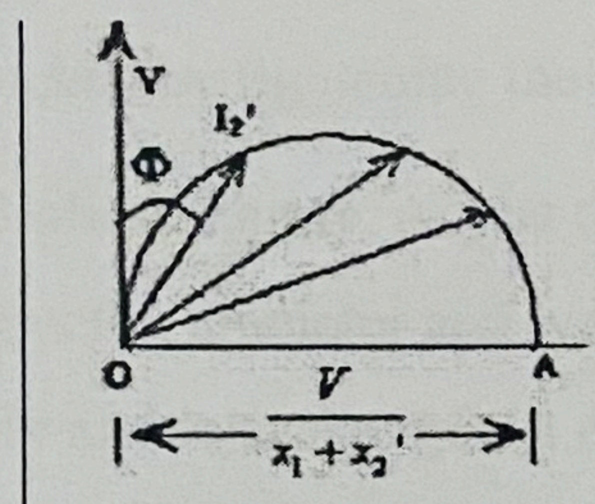

Figure 1

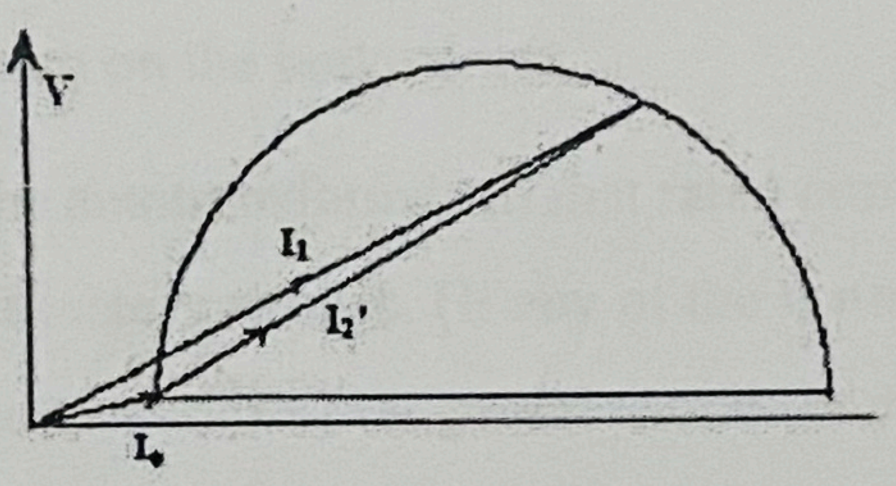

Figure 2

Thus in the case of induction motors, the locus of the current due to load lies on a circle and the diagram is known as a circle diagram. If no load current taken by the motor is also to be accounted for to obtain the stator current, the diagram can then be shown as in figure 2. The stator current I_1 is then the phasor sum of I_2' and I_0

No load and blocked rotor tests are conducted for determining the equivalent circuit parameters, for predetermining the efficiency at any load and to draw the circle diagram. No-load test is conducted at rated voltage keeping the motor on no-load. Since the no-load current is only 20-40% of the full load current, the copper losses can be neglected. Input power is equal to constant iron, friction and windage losses of the motor. In blocked rotor test, rotor is blocked and a reduced voltage is applied to the stator through a 3-phase autotransformcr. Due to low voltage and no rotation, core and mechanical losses are neglected. Input power is equal to copper loss only.

PROCEDURE

A. NO-LOAD TEST

Make the connections as shown in figure.

Precautions:

i) Keep the autotransformer in minimum voltage position

ii) Keep belt on brake drum in loose position (motor on no load)